

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0315	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
Student Name	K SAMUEL RAJ	Reg. No.	1913260140
Section/Batch	Slot D	Date	03-10-2026

Code of Conduct:

I, K SAMUEL RAJ, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

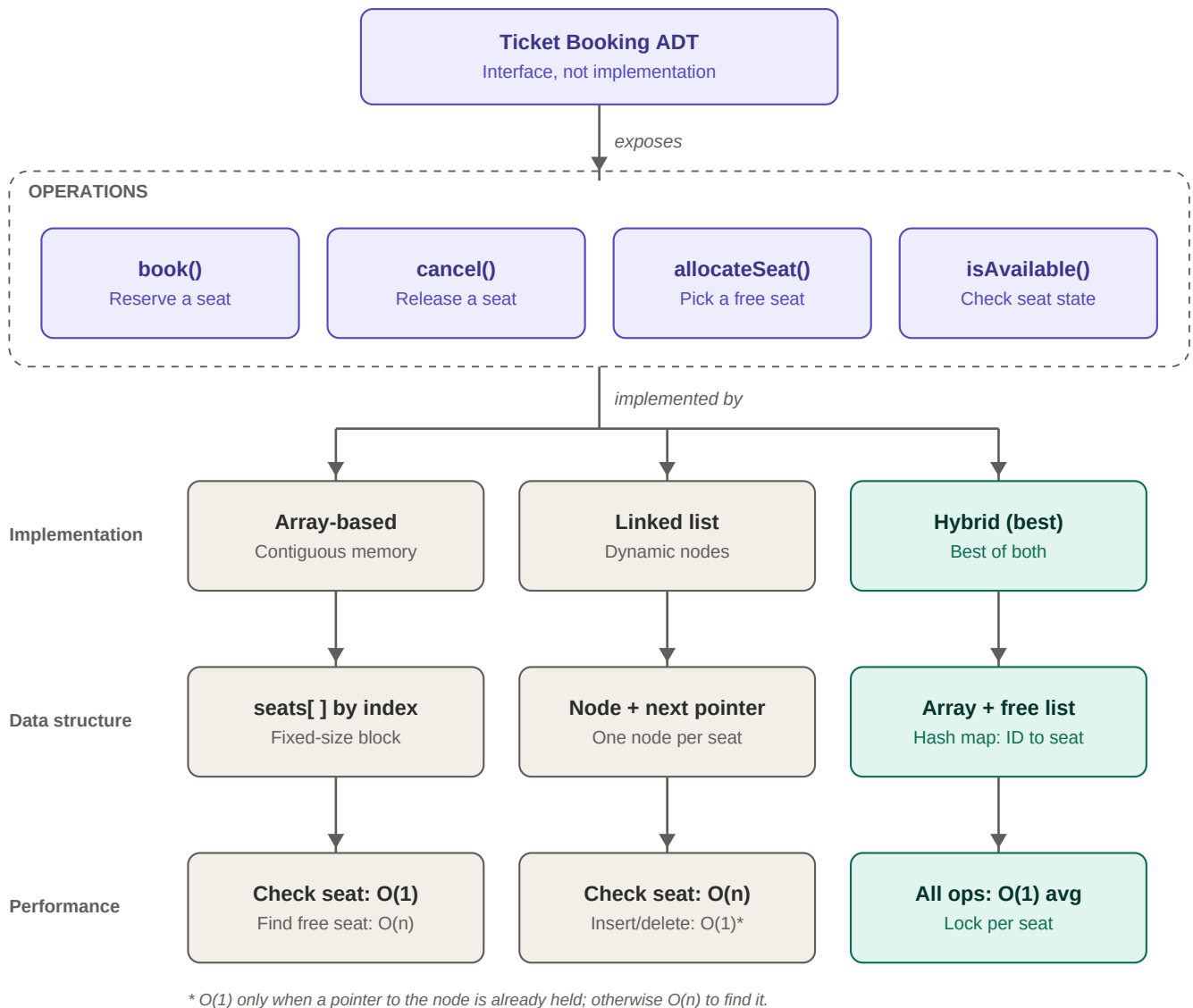
Signature of the student:

K Samuel Raj

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

Ticket Booking System: Concept Map

ADT → Operations → Implementation → Data Structures → Performance



Array: $O(1)$ lookup by seat number and cache-friendly, but finding a free seat needs an $O(n)$ scan and capacity is fixed.

Linked list: flexible size and cheap insert/delete, but every lookup is $O(n)$ and pointers add overhead.

Best for real-time booking: the hybrid. Seat numbers are fixed and addressed directly, so the array's weakness rarely matters. A free list gives $O(1)$ allocation, a hash map gives $O(1)$ cancellation, and a per-seat lock prevents double booking.